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Self-Supervised Ensemble Learning for Skin Lesion Classification with Limited Annotated Data

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A Thesis submitted for the degree of Bachelor of Science (BSc)in
Computer Science and Engineering (CSE) at
American International University Bangladesh (AIUB)
Faculty of Science and Technology (FST)

Summer 2024-2025 Semester
Submission Date: March 7 ,2026

Declaration

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Acknowledgement

The authors would like to express their sincere gratitude to ( ) for precious guidance and continuous support throughout this research. Colleagues' and peers' inputs in providing constructive criticism are also gratefully accepted. Finally, the authors appreciate their family members for continuous support while undertaking this work.

Author Contributions

List the significant and substantial inputs made by different authors to this research, work and writingrepresented and/or reported in the thesis. These could include significant contributions to the conception and design of the project; non-routine technical work; analysis and interpretation of researchdata; drafting significant parts of the work or critically revising it to contribute to the interpretation.

	Anzoom, Tanvir	Adir, Md Sarafat Ali	Mahatab, Sharif Hadi	Rifa, Tasnim Zaman		**Comments**
	20-41920-1	***20-41926-1***	***20-43625-2***	***21-45663-3***		
0 - 3 points	**Perform as effective individual**					
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Reflection on feedback	3	3	3	3		
Quality of work	3	3	3	3		
Self-directed	3	3	3	3		
0 - 3 points	**Perform as effective team member/leader**					
Taking responsibility	3	3	3	3		
Contribution	3	3	3	3		
Collaboration	3	3	3	3		
Working with others	3	3	3	3		
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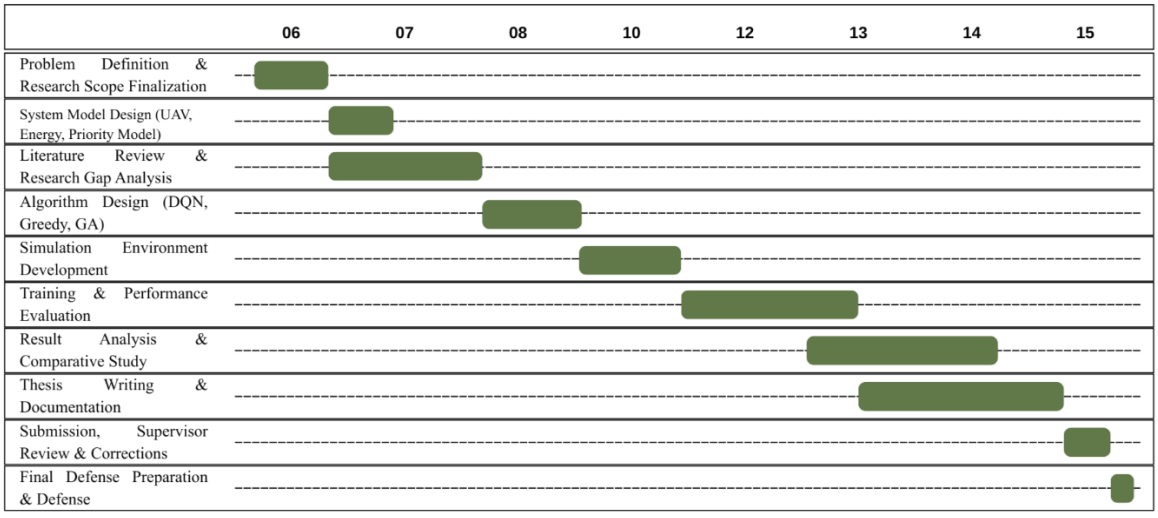
Presentation delivery	3	3	3	3		
Voice and tone	3	3	3	3		
Enthusiasm	3	3	3	3		
Creativity & Tools use	3	3	3	3		

Project-Thesis Planning

Project Tasks	Schedule Data	Execution Data
1. Planning	June 06, 2025	June 06, 2025
2. Literature Review	June 13, 2025	June 15, 2025
3. Survey question selection and review	June 20, 2025	June 22, 2025
4. Conduct survey/interview	August 18, 2025	August 20, 2025
5. Writing thesis report	December 18, 2025	December 23, 2025
6. Submission and review	February 27, 2026	March 7, 2026



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List of Abbreviations

Abbreviation	Full Meaning
AI	Artificial Intelligence
ANN	Artificial Neural Network
BCC	Basal Cell Carcinoma
CNN	Convolutional Neural Network
CS	Computer Science
CSE	Computer Science and Engineering
D-Features	Dimensional Features
FST	Faculty of Science and Technology
FT	Fine-Tuning
GAN	Generative Adversarial Network
GAP	Global Average Pooling
GPU	Graphics Processing Unit
HCI	Human Computer Interaction
ISIC	International Skin Imaging Collaboration
KNN	K-Nearest Neighbor
ML	Machine Learning
NLP	Natural Language Processing
NN	Neural Network
ReLU	Rectified Linear Unit
RNN	Recurrent Neural Network
ROC	Receiver Operating Characteristic
RQ	Research Question
SCC	Squamous Cell Carcinoma
SGD	Stochastic Gradient Descent
SimCLR	Simple Framework for Contrastive Learning
SSL	Self-Supervised Learning
WBS	Work Breakdown Structure

Abstract

The automatic diagnostic systems need extensive expert-labeled data which costs money and presents challenges for acquisition. The research develops a hybrid Self-Supervised Ensemble Learning framework to solve the "data bottleneck" issue which affects medical imaging skin lesion classification. The research investigates if combining self-supervised representation learning with ensemble methods enables the system to maintain accurate diagnostic results while operating with limited labeled data.

The research uses the HAM10000 dataset which contains 10015 dermoscopic images that span across seven different categories. The main methodology consists of a two-step training process. The first step uses SimCLR (Simple Framework for Contrastive Learning) to pretrain a ResNet50 backbone on unlabeled data for domain-specific feature learning without needing human-made annotations. The pretrained encoder gets combined with a supervised ResNet50 and a lightweight EfficientNetB0 model to create an ensemble system. The models needed to be fine-tuned using only 20 percent of the available labeled data because the study wanted to create a realistic scenario which had restricted label availability. The three architectures' predictions go through probability-level soft-voting which leads to a final diagnostic output.

The research creates an important solution which helps healthcare systems that need to conduct image labeling work but lack the funds for extensive manual operations. The model develops the ability to recognize advanced visual characteristics in skin lesions through self-supervision training which includes identifying texture and color pattern differences, which standard supervised models fail to detect when they have small data sets.

The experimental results demonstrate that the proposed ensemble framework achieves an overall test accuracy of 76% and a macro-average ROC-AUC of 0.914. The results show substantial improvements in all baseline models which serve as comparison points. The ensemble method demonstrated better performance at detecting minority classes which included vascular lesions and dermatofibroma while it achieved an AUC score above 0.86 across all lesion types.

The research findings produce two main effects. The scientific study shows that self-supervised pretraining results in better medical image classifier performance than standard ImageNet initialization. The system delivers a data-efficient diagnostic solution which enables deployment in settings where dermatopathology expertise remains absent. This framework advances the field of medical AI by proving that high-performance, reliable diagnostic aids can be built with significantly reduced human annotation effort which leads to faster and cheaper clinical decision support systems.

Chapter 1
Introduction

1.1 Background Analysis

Medical literature states that aberrant skin cell proliferation, usually brought on by unchecked division because of DNA damage, is a hallmark of skin cancer (Abbas et al., 2021). The detection of cancerous tumors needs to occur before their diagnosis because thermoscopic examinations depend on expert knowledge and personal judgment for manual evaluation. Recent advances in deep learning show to achieve remarkable outcomes through dermatological picture analysis which uses convolutional and transformer-based networks for visual recognition tasks. The conventional method which trains these models requires extensive amounts of accurately marked information. Medical professionals need dermatopathological verification before any annotation work can take place, which makes this process both costly and difficult to implement in the medical field. The self-supervised learning method enables researchers to develop image representations that function across different tasks by completing proxy tasks with complete access to unmarked data. The testing phase allows models to undergo fine-tuning by using a small number of marked data points, which enables their performance assessment against fully supervised models.
The combination of self-supervised learning with ensemble learning, which combines multiple predictions for better accuracy and reduced uncertainty, provides additional advantages through enhanced model performance and improved efficiency of confidence score determination. The combination of these techniques establishes a new method for skin lesion classification which functions accurately despite having limited available annotations.

1.2 Existing Studies

Several studies have explored automated skin lesion classification using deep learning and ensemble methods. For example, **Park et al. (2021)** proposed a *deep ensemble learning framework* that combined multiple CNN models for skin lesion classification and achieved higher accuracy compared to single CNNs. However, their method relied entirely on large, annotated datasets, which are not always available in clinical practice (Park et al., 2021).
Similarly, **Ahsan et al. (2020)** used *transfer learning with CNN architectures* such as ResNet and Dense Net for skin cancer detection. While they obtained good performance, their study was limited by dependency on extensive labeled datasets (Ahsan et al., 2020).
Reyes et al. (2023) emphasized the potential of *self-supervised learning (SSL)* in medical image classification, showing that SSL can learn rich image representations even with fewer labels. However, they focused on a single model setup and did not explore ensemble integration (Reyes et al., 2023).
Recently, **Zhao et al. (2024)** proposed a *SimCLR-based SSL framework* for medical imaging and proved that it performs better than supervised models when annotated data is limited. Yet, their research did not combine SSL with ensemble learning for improved results (Zhao et al., 2024).
Moreover, **Hassan et al. (2024)** discussed a *multi-level ensemble model* that used different CNNs with attention mechanisms to improve skin lesion detection accuracy. Although the performance was impressive, the system still required large amounts of labeled data (Hassan et al., 2024).
Finally, **Manivannan (2025)** introduced a *semi-supervised online knowledge distillation*

framework for skin lesion classification, highlighting that limited-label settings can be improved using teacher-student ensembles, but this still lacked a full self-supervised ensemble structure (Manivannan, 2025).

Research Gap

Based on our studies, it is evident that most previous research either focused on fully supervised ensemble models requiring large, labeled datasets or explored self-supervised learning independently. Very few studies have attempted to combine self-supervised and supervised models into an ensemble framework for skin lesion classification under limited annotated data conditions, a challenge further complicated by the high intra-class variance in medical imaging (Li et al., 2023).

How This Study Addresses the Gap
This research closes the gap by developing a Self-Supervised Ensemble Learning model that integrates:

- ResNet50 (Supervised) – for labeled data learning,
- ResNet50 (SSL - SimCLR) – for representation learning from unlabeled data, and
- EfficientNetB0 – for lightweight, efficient feature extraction.

By combining these models through an ensemble approach, the proposed study reduces dependence on large, labeled datasets while improving classification accuracy and robustness extending previous work and providing a more practical solution for real-world skin cancer diagnosis.

1.3 Research Motivation and Objective

Through the development of a self-supervised ensemble learning system, this study seeks to improve skin lesion classification performance in situations where annotated data is scarce. The objective is to lessen reliance on big labeled datasets by combining the advantages of ensemble learning for robustness and self-supervised learning (SSL) for representation learning.(Reyes, Li, & Zhang, 2023; Zhao, Lin, & Zhang, 2024).Previous studies using CNN ensembles (Park, Lee, & Kim, 2021) or transfer learning (Ahsan, Shafique, & Manzoor, 2020) achieved high accuracy but relied heavily on extensive labeling. Others explored SSL individually without combining it with ensembles (Reyes et al., 2023; Zhao et al., 2024). Hence, a gap exists in combining supervised and self-supervised models within a single ensemble for data-efficient medical image classification.
This study addresses that limitation by:

1. Developing a hybrid ensemble combining ResNet50 (Supervised), ResNet50 (SSL – SimCLR), and EfficientNetB0.
2. Evaluating performance on limited annotated datasets to assess accuracy, precision, and F1-score.
3. Demonstrating that self-supervised ensembles can provide reliable diagnostic tools with reduced data requirements.

Research Questions

- **RQ1:** How can self-supervised learning (SimCLR) improve skin lesion classification with limited annotated data?
- **RQ2:** What is the comparative performance of ResNet50 (Supervised), ResNet50 (SSL–SimCLR), and EfficientNetB0?
- **RQ3:** Can combining supervised SSL models in an ensemble yield higher accuracy and robustness than single models?

Research Contribution

Dermatologists, medical imaging researchers, and healthcare organizations that depend on automated skin lesion analysis are the research's target users. The Self-Supervised Ensemble Learning Framework that has been suggested will assist:

- Dermatologists by offering a diagnostic aid that improves the precision of early detection.
- Researchers provide a novel hybrid framework for limited-data contexts that combines SSL and ensemble learning.
- Healthcare systems by lowering the requirement for big datasets with expert annotations, allowing for quicker and more economical deployment.

This thesis advances scientific knowledge in the classification of medical images by:

- Presenting a hybrid ensemble technique that combines self-supervised (SimCLR) and supervised (ResNet50) learning for performance that is data efficient.
- proving that, even with minimal annotation, accuracy and generalization are enhanced by the ensemble fusion of SSL and conventional models.
- presenting an empirical investigation of skin lesion categorization that closes the gap between ensemble-based optimization and representation learning.

Chapter 2

Research Methodology

2.1 Research Methodology

This chapter illustrates the methodological framework used to develop a self-supervised ensemble learning model for skin lesion classification using the HAM10000 dataset, which serves as a benchmark for comparing human diagnostic accuracy against machine-learning algorithms (Tschandl et al., 2019), making it one of the largest publicly available dermoscopic image databases, is discussed. The proposed approach aims to address the challenge of the scarcity of labeled medical data by using the self-supervised contrastive learning method in conjunction with the supervised fine-tuning of the ensemble of deep convolutional neural networks.

2.1.1 Dataset Description

The experiments are performed using the HAM10000 dermoscopic image dataset, which is defined as "Human Against Machine." This is one of the largest publicly available dermoscopic images used for automated skin lesion analysis. This dataset consists of a total of 10,015 high-resolution RGB dermoscopic images, including seven clinically validated diagnostic classes, namely, actinic keratoses (akiec), basal cell carcinoma (bcc), benign keratosis (bkl), dermatofibroma (df), melanoma (mel), melanocytic nevi (nv), and vascular lesions (vasc).

A substantial class imbalance characterizes the dataset. The melanocytic nevi (nv) category accounts for more than 60% of the total samples, whereas minority classes such as dermatofibroma (df) and vascular lesions (vasc) contain fewer than 150 instances each. Such imbalance presents a significant challenge for supervised learning models, as it may bias predictions toward majority classes and degrade generalization on rare lesion types.

To mitigate these limitations, his study proposes a novel approach that incorporates self-supervised contrastive representation learning, along with class-weighted loss functions in the process of supervised fine-tuning. The class distribution of the dataset is shown in Figure 2.1.

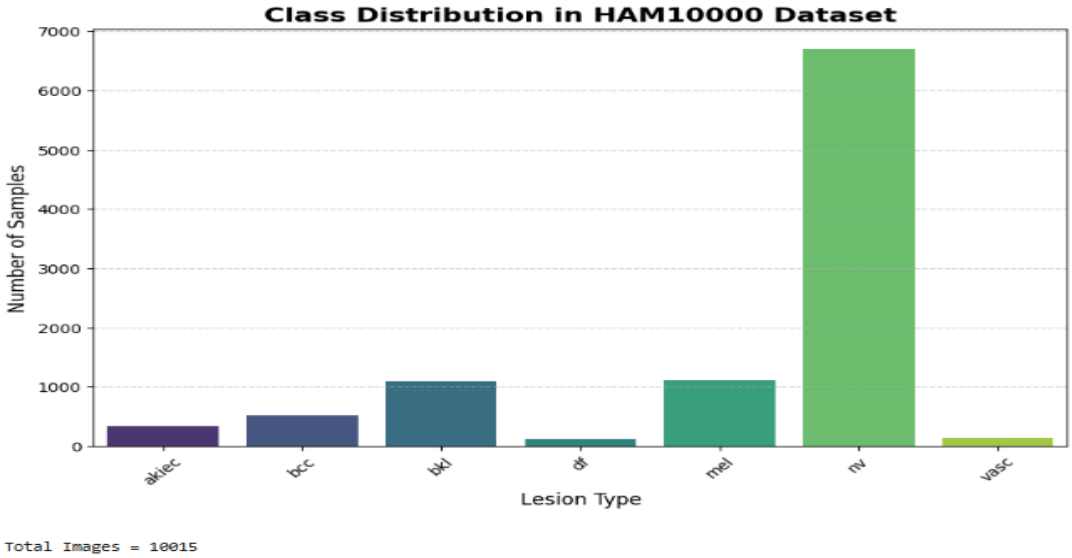


Figure 2.1: Class Distribution of the HAM10000 Dataset

2.1.2 Experimental Setup

The experiments were all performed under a GPU-accelerated cloud environment, which uses an NVIDIA Tesla T4 GPU with 16 GB memory. The implementation is written in Python 3.10, which uses TensorFlow 2.x and the Keras deep learning API.

The input images for dermoscopy were resized to a size of 224 x 224 x 3 to ensure compatibility with standard convolutional neural network architectures. Optimization of the models was performed using the Adam optimizer with an initial learning rate of 1×10^{-4} and batch size of 32 for optimization of the models. Early stopping of the models was performed to prevent overfitting of the models. Class-weighted loss functions were incorporated during the training of the models.

All experiments were conducted under identical computational settings to ensure reproducibility and fair performance comparison among individual classifiers and ensemble configurations.

2.2 Procedures

The process used in the study involves dataset organization/preprocessing, data transformation for supervised learning and self-supervised learning, model architecture development, SimCLR contrastive pretraining, CNN classifiers, supervised fine-tuning, and ensemble model integration. The process also involves training the model using Adam optimization, regularization, early stopping, testing the model using accuracy, F1-score, ROC AUC, etc. The researchers monitored all the processes to get the best model performance when the labels of only 20% of the dataset are available.

The overall workflow is illustrated in **Figure 2.2**, which provides a high-level overview of the proposed research methodology under a low-label (20%) supervision setting.

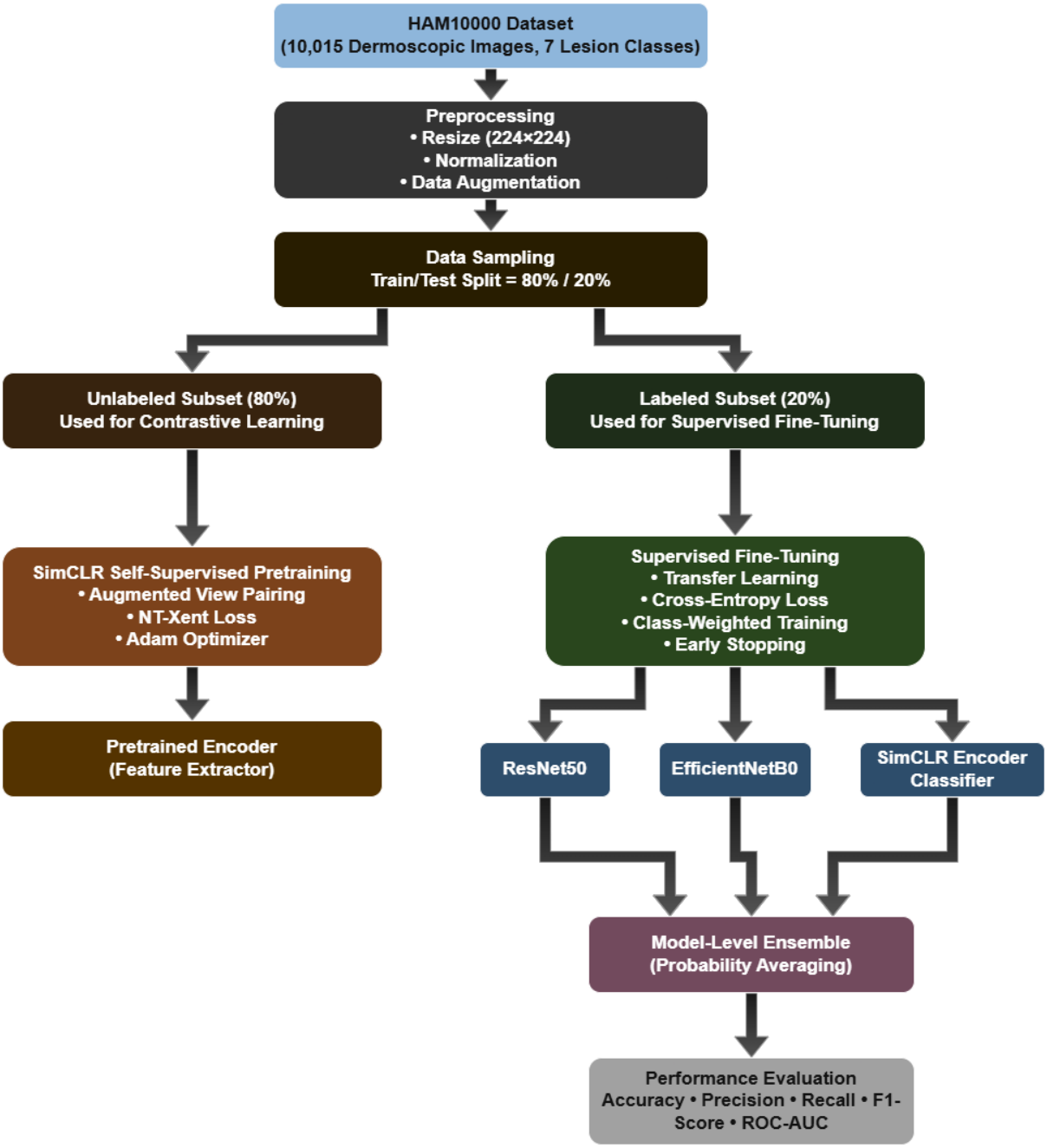


Figure 2.2: High-Level Workflow of the Proposed Research Methodology

2.2.1 Data Preparation

The dataset used in this study is the HAM10000 ("Human Against Machine with 10,000 training images") dermoscopic image dataset, which includes a total of 10,015 images in the following seven clinically significant skin lesion categories: akiec, bcc, bkl, df, mel, nv, and vasc.The dataset includes RGB dermoscopic images, which show different levels of resolution and different types of imaging conditions.

To ensure uniformity, all the images are resized to a dimension of 224 x 224 pixels, as per the specifications of the deep learning models, namely ResNet50 and EfficientNetB0. The dataset

is derived from the ISIC Archive, and researchers are expected to adhere to the licensing terms and ethical guidelines as recommended by the ISIC Archive for medical image usage.To ensure effective supervised and self-supervised learning, the dataset is split into three parts using stratified sampling, i.e., 80% for the Training Set, 10% for the Validation Set, and the remaining 20% for the Test Set.In the study, semi-supervised learning is considered, which is less than fully annotated and requires only 20% of the training data for supervised learning. The remaining training data is utilized for contrastive pretraining using the SimCLR method. The process begins with file paths and labels, which are mapped after splitting the data, and duplication is removed to ensure consistent handling of the data and avoid the possibility of data leakage.

2.2.2 Model Architecture

The proposed architecture incorporates supervised convolutional neural networks and a self-supervised contrastive encoder for improved classification performance under limited labeled data settings. Three backbone networks were used, namely, ResNet50, EfficientNetB0, and a SimCLR pretrained ResNet50 encoder.

Each input image of size $224 \times 224 \times 3$ is processed through one of the pretrained convolutional backbones to extract hierarchical feature representations.

The ResNet50 is a deep residual network with 50 layers, and it uses identity skip connections for the purpose of gradient propagation and deep feature learning. The EfficientNetB0 uses a compound scaling approach for achieving parameter efficiency and strong performance by balancing depth, width, and resolution.

For the self-supervised branch, a ResNet50 backbone was pretrained using the SimCLR framework with the NT-Xent contrastive loss. In the pretraining stage, the projection head was added to the encoder, which was removed during the classification stage, and the encoder was utilized in the supervised stage.

Following the extraction of backbone features, Global Average Pooling (GAP) was used to map the spatial feature maps to concise feature vectors. The Dropout layer with a dropout rate of 0.25 was used to prevent overfitting. Finally, the fully connected dense layer with softmax activation generated the probability distribution over the seven classes of lesions.

The outputs of the three classifiers were subsequently combined through a probability-based ensemble strategy to enhance robustness and predictive stability.

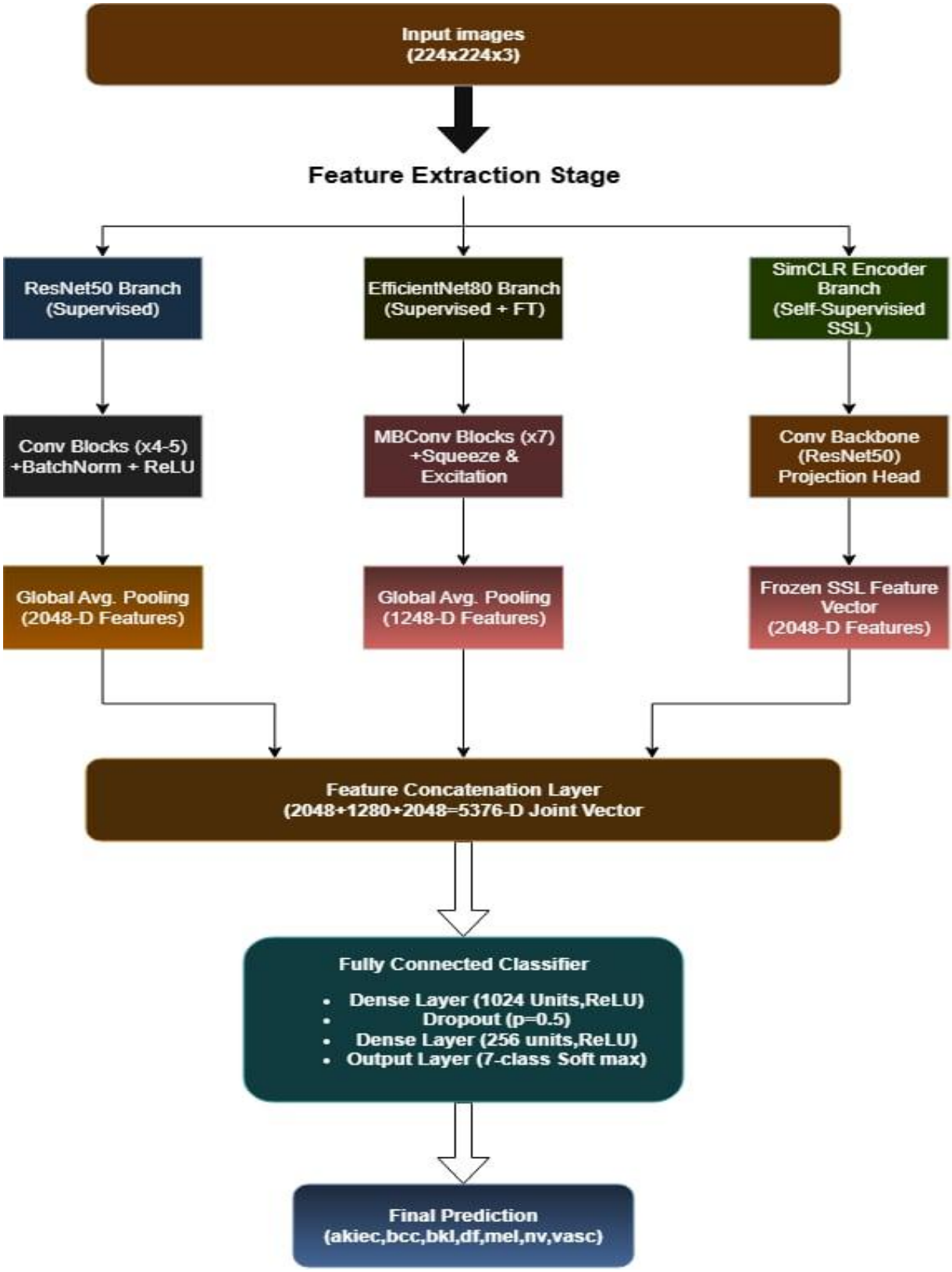


Figure 2.3: Proposed Hybrid Self-Supervised Ensemble Architecture

2.2.3 Data Transformation

Data transformation plays a critical role in improving the robustness, generalization, and representation-learning capability of the proposed ensemble model. In this study, the use of supervised augmentation for labeled data and contrastive augmentation for SimCLR-based self-supervised learning is proposed. These data transformations are vital in addressing issues of class imbalance and overfitting, especially under limited labeled data conditions.

Supervised Data Augmentation

For the supervised learning approach, moderate augmentations have been employed for ResNet50 and EfficientNetB0. These augmentations simulate the variability seen in dermoscopic imaging, which includes flipping, small rotation, changes in brightness, contrast, and zooming. These augmentations will enable the model to be invariant to changes in illumination, orientation, and small changes in shape, which is usually seen in skin lesion imaging.

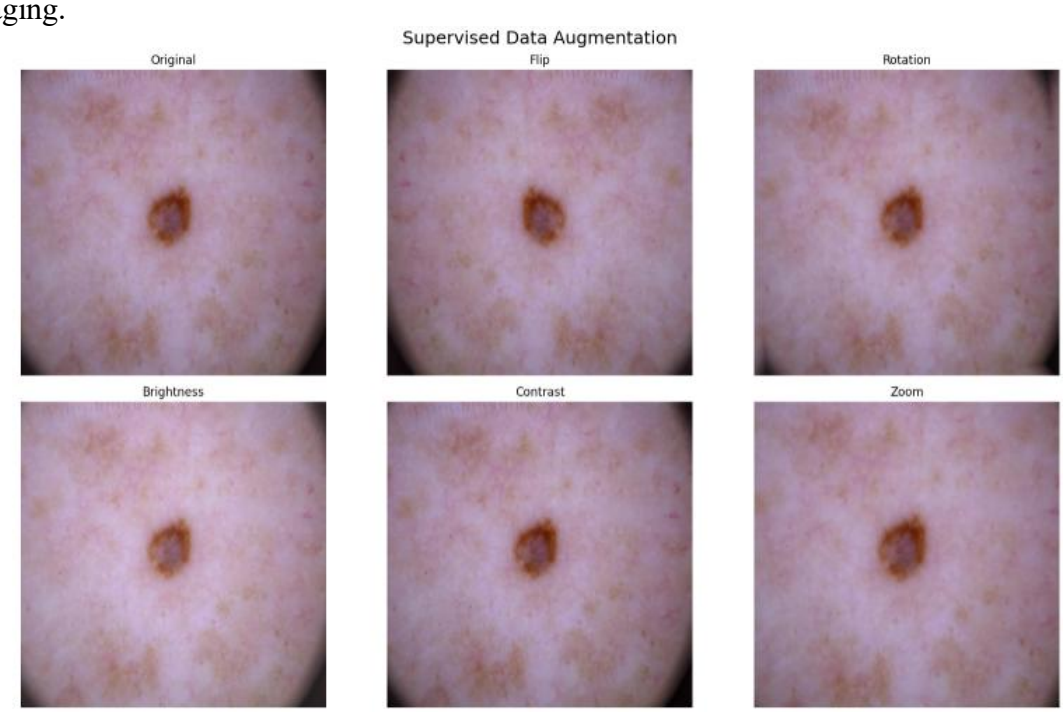


Figure 2.4: Comparison of Moderate Supervised Data Augmentation Techniques

Self-Supervised (SimCLR) Data Augmentation

In the pretraining step of the SimCLR, the augmentation pipeline was enhanced to produce two stochastic views of the same image. The goal is to increase the agreement between the representation of the two views of the same image, while at the same time increasing the difference between the representation of two different images.

The contrastive augmentation process involved the following transformations: random resized crops with a scale range of 0.6 to 1.0, horizontal flip, brightness perturbation, and contrast adjustment. These strong transformations force the network to learn invariant and semantically meaningful feature representations, regardless of the absence of any class label information.

Such invariant representations greatly improve the overall supervised performance when the availability of labeled data is low.

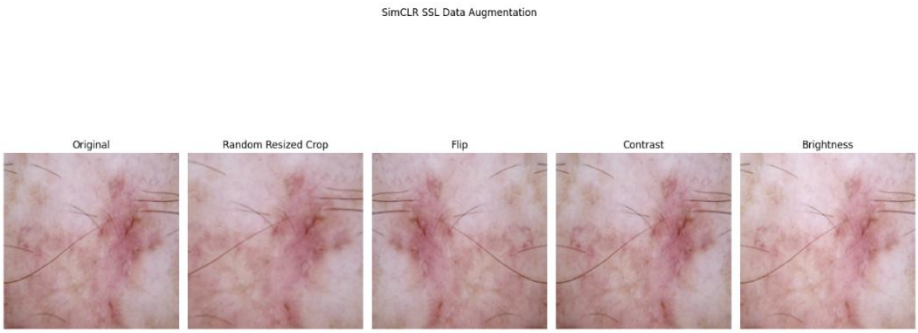


Figure 2.5: Visualization of Strong SimCLR Contrastive Augmentation Pipeline

2.2.4 Model Training

The training process, as employed in the current study, can be divided into two main stages: the self-supervised pretraining via the SimCLR method, followed by the supervised fine-tuning of the classifiers. The rationale behind the two-stage training process is to effectively utilize the vast number of unlabeled dermoscopic images for the learning of generalizable feature representations.

Self-Supervised Pretraining (SimCLR)

In the first stage, the ResNet50 model is utilized as the backbone network, which is trained under the SimCLR framework. For an input image x, two stochastic augmentations $t_1(x)$and $t_2(x)$are generated using a composition of random cropping, color jittering, flipping, and Gaussian blur. These augmented views are passed through the encoder $f(\cdot)$followed by a projection head $g(\cdot)$to obtain latent representations:

$$zi = g(f(t1(x))), zj = g\left(f\left(t2(x)\right)\right)$$

The training objective maximizes agreement between representations of positive pairs while minimizing similarity to other samples within the batch. The normalized temperature-scaled cross-entropy (NT-Xent) loss for a positive pair $(i, j)$is defined as:

$$\ell i,j = -log\sum k = 12N1[\,k \neq i\,]exp\left(\frac{sim(zi,zk)}{\tau}\right)exp\left(\frac{sim(zi,zj)}{\tau}\right)$$

where $\mathrm{sim}(z_i, z_j) = \frac{z_i^\top z_j}{\|z_i\|\|z_j\|}$denotes cosine similarity, $\tau$is the temperature scaling parameter, and Nrepresents the batch size. The indicator function $\mathbf{1}_{[\,k\neq i\,]}$ensures that the denominator excludes the anchor sample itself.

The goal of this objective is to encourage the alignment of positive pairs, as well as the uniformity of the negative pairs in the representation space. Optimizing this loss on unlabeled dermoscopic images allows the encoder to learn domain-invariant and semantically meaningful features.

The pretraining is conducted for 30 epochs using the Adam optimizer with the learning rate set to 1×10^(-4). This step only uses unlabeled training data, allowing for scalable feature learning under limited supervision.

Supervised Fine-Tuning

Following self-supervised pretraining, the learned encoder weights are transferred to a classification head for supervised optimization at the same time, two other models, namely ResNet50 and EfficientNetB0 with pre-trained weights on ImageNet, are trained independently using only 20% of the labeled dataset from HAM10000.
Let $y \in \{1, \dots, C\}$ denote the true class label and $\hat{p}_c$ represent the predicted probability for class c. The supervised objective minimizes the sparse categorical cross-entropy loss:

$$\mathcal{L}_{sup} = -\sum_{c=1}^{C} y_c \log(\hat{p}_c)$$

where C is the total number of lesion categories. To mitigate class imbalance, class-weighting is applied such that underrepresented categories receive higher penalty weights during optimization.

All the models are fine-tuned using the Adam optimizer with a learning rate of 1×10^(-4) for 10 epochs. Some of the deeper backbone layer parameters are kept unfrozen to support domain adaptation, while avoiding the catastrophic forgetting of the pre-trained features. The batch size is set to 32, and on-the-fly data augmentation is used to boost the generalization.

To improve optimization stability, the ReduceLROnPlateau scheduler dynamically reduces the learning rate when validation loss stagnates. Model performance is monitored on a held-out validation set after each epoch.

Training Configuration and Optimization Strategy

The training process uses a batch size of 32, and data augmentation is done on the flight to ensure better generalization. The validation set is utilized for checking the training progress after each epoch. For better stability during the training process, ReduceLROnPlateau is utilized to adjust the learning rate if the validation loss plateaus. Early stopping is done with patience to avoid overfitting and retain the best weights.

Ensemble Model Formation

Upon completion of individual training, predictions from the three classifiers are combined through probability-level averaging. Let $p^{(1)}, p^{(2)}, p^{(3)}$ denote the predicted class probability vectors from ResNet50, EfficientNetB0, and the SimCLR-based classifier, respectively. The final ensemble prediction is computed as:

$$p^{ens} = \frac{1}{3}\left(p^{(1)} + p^{(2)} + p^{(3)}\right)$$

The final predicted class is determined by:

$$\hat{y} = \arg\max_{c} \; p_c^{ens}$$

This soft-voting mechanism reduces variance, mitigates model-specific bias, and leverages complementary feature representations across architectures, thereby enhancing classification robustness.

2.2.5 Early Stopping

To avoid overfitting and ensure the stability of the model's convergence, an early stopping method is used. In this method, the validation loss is monitored after the end of each epoch, and the model is stopped if there is no improvement in the validation loss over a set number of epochs, referred to as patience. In this study, the patience was set to 5, implying that if there is no improvement in the validation loss after five epochs, the model is stopped.

Early stopping is applied individually to all three models—ResNet50, EfficientNetB0, and the SimCLR-based classifier—during supervised fine-tuning. When the validation loss of the model is minimal, the weight of the model is saved as the best model so that the final model is the optimal one in terms of generalization rather than the last one, which might be subjected to overfitting.

As part of this experimental process, each model had a unique stopping point. This is illustrated by the fact that, for instance, the validation loss for the SimCLR-based classifier reached its minimum earlier than it did for the fully supervised classifiers, after which it plateaued. This meant that, since the validation loss was no longer improving, it was time to apply early stopping, retaining the best-performing model. This is equally true for the ResNet50 and EfficientNetB0 models. By applying this early stop, it is possible to improve the overall robustness and stability of the classification system.

2.2.6 Validation Methodology

The validation methodology is intended to evaluate the performance, reliability, and generalization ability of the proposed SSL-Supervised Ensemble approach for skin lesion classification. The evaluation is based on the extent to which the trained models can classify the seven skin lesion classes from the HAM10000 dataset under unknown test conditions. The validation process is intended to ensure the accuracy, reliability, and significance of the ensemble's prediction results.

Model Initialization and Loading

As part of the model validation process, the three trained models, namely, ResNet50 (supervised), EfficientNetB0 (supervised), and the SimCLR pretrained ResNet50 classifier, are loaded in evaluation mode. The weight checkpoints used in this process are the best performing versions of the models, obtained using the early stopping technique.

Data Preprocessing

All the images used in the test set went through a preprocessing routine that is similar to the training procedure. This involved resizing the images to 224 x 224 pixels, converting the images to float32 type, and normalizing the input images according to the respective preprocessing methods used in the ResNet50 and EfficientNet models. There was no use of data augmentation in the validation procedure, which helped maintain the original clinical features of the dermoscopic images.

Inference and Ensemble Prediction

During the inference stage, each test image was individually fed into the three trained classifiers, namely the supervised ResNet50, the supervised EfficientNetB0, and the SimCLR-pretrained ResNet50 classifier. Each classifier generated hierarchical feature representation and produced the posterior probability distribution over the seven skin lesion classes through the softmax activation function.

Instead of relying on the output of one model, an ensemble-based soft voting approach was utilized for the stability of the prediction. For this purpose, the class probability vectors of the three networks were averaged based on the predicted probability for each class. The resulting probability distribution is the ensemble system's prediction.
The final class label was determined based on the class for which the highest value exists in the averaged probability distribution.

Through the combination of various architecture and learning paradigm predictions, such as supervised and self-supervised learning, the ensemble method minimizes model-specific bias and prediction variance. This enables better generalization capability and greater robustness, especially when dealing with minority lesion categories that are visually ambiguous.

Qualitative Evaluation

To complement quantitative metrics, a subset of test samples was randomly selected for qualitative inspection for each test sample, the ground truth label, predicted label, and corresponding confidence were analyzed. This approach helped in the identification of systematic errors and class problems, especially for the underrepresented classes such as akiec, df, and vasc.

Quantitative Performance Metrics

Model performance was assessed using a comprehensive set of classification metrics:

- **Accuracy:** Overall proportion of correctly classified samples.
- **Precision, Recall, and F1-score:** Computed per class and macro-averaged to account for class imbalance.
- **Confusion Matrix:** Used to analyze inter-class misclassification trends.
- **Receiver Operating Characteristic (ROC) curves and Area Under the Curve (AUC):** Calculated in a one-vs-rest manner for each class to evaluate discriminative capability across varying decision thresholds.

2.3 Systematic Literature Review (SLR)

The objective of this SLR study was to assess both self-supervised learning (SSL) and ensemble methods as they currently exist for dermatological image classification needs. The study evaluated performance outcomes which occurred under conditions of limited data availability.

REFERENCE	METHODOLOGY FOCUS	MODELS USED	DATA STRATEGY	KEY PERFORMANCE / FINDINGS
TSCHANDL ET AL. (2019)	DATASET BENCHMARKING	MACHINE LEARNING VS. HUMANS	HAM10000 DATASET	ESTABLISHED THE HAM10000 DATASET AS A GOLD STANDARD FOR COMPARING AI AND HUMAN DIAGNOSTIC ACCURACY.
AHSAN ET AL. (2020)	TRANSFER LEARNING	RESNET, DENSENET	100% LABELED	OBTAINED GOOD PERFORMANCE BUT REMAINS LIMITED BY HEAVY DEPENDENCY ON EXTENSIVE LABELED DATASETS.
CHEN ET AL. (2021)	SELF-SUPERVISED (SIMCLR)	RESNET50 (BACKBONE)	UNLABELED PRETRAINING	INTRODUCED THE SIMCLR FRAMEWORK, PROVING THAT CONTRASTIVE LEARNING CAN LEARN POWERFUL REPRESENTATIONS WITHOUT LABELS.
PARK ET AL. (2021)	SUPERVISED ENSEMBLES	CNNS (RESNET, VGG)	100% LABELED	ACHIEVED HIGH ACCURACY BUT LACKS DATA EFFICIENCY REQUIRED FOR CLINICAL SETTINGS WITH SPARSE ANNOTATIONS.
AZIZI ET AL. (2021)	SEMI-SUPERVISED LEARNING	BIG SSL MODELS	CROSS-DOMAIN SSL	DEMONSTRATED THAT SELF-SUPERVISED PRETRAINING SIGNIFICANTLY OUTPERFORMS STANDARD IMAGENET INITIALIZATION FOR MEDICAL TASKS.
REYES ET AL. (2023)	SELF-SUPERVISED (SSL)	RESNET50	10% - 50% LABELED	PROVED THAT SSL CAN LEARN RICH FEATURES EVEN WITH FEWER LABELS, THOUGH LACKED ENSEMBLE INTEGRATION.
LI ET AL. (2023)	CONTRASTIVE LEARNING	CNN-BASED SSL	LABEL-EFFICIENT	SHOWED THAT CONTRASTIVE LEARNING EFFECTIVELY HANDLES HIGH INTRA-CLASS VARIANCE IN SKIN LESION CLASSIFICATION.
ZHAO ET AL. (2024)	CONTRASTIVE LEARNING	SIMCLR	LIMITED LABELED	CONFIRMED SIMCLR OUTPERFORMS SUPERVISED IMAGENET PRETRAINING WHEN ANNOTATED DATA IS SCARCE.
HASSAN ET AL. (2024)	ATTENTION MECHANISMS	CNN + ATTENTION	100% LABELED	IMPROVED ACCURACY IN MINORITY CLASSES BUT STILL REQUIRED LARGE AMOUNTS OF EXPERT-LABELED DATA.
MANIVANNAN (2025)	KNOWLEDGE DISTILLATION	TEACHER-STUDENT	SEMI-SUPERVISED	ACHIEVED GOOD RESULTS IN LIMITED-LABEL SETTINGS BUT LACKED A FULL SELF-SUPERVISED ENSEMBLE STRUCTURE.
OUR STUDY (2026)	HYBRID SSL-ENSEMBLE	RESNET50, SIMCLR, EFFICIENTNET	20% LABELED	ACHIEVED 76% ACC & 0.91 AUC VIA MULTI-MODEL FUSION, SPECIFICALLY MITIGATING THE 20% LABEL CONSTRAINT.

Table 2.2: Systematic Literature Review (SLR) of Deep Learning and SSL in Dermatology

This table highlights the specific focus areas, architectures, and performance outcomes of previous studies compared to this research.

The SLR confirmed that while supervised ensembles (Park et al.) and independent SSL models (Zhao et al.) are well-documented, there is a distinct lack of research integrating a SimCLR-pretrained self-supervised encoder into a heterogeneous ensemble (mixing different CNN families) to specifically mitigate the 20% label constraint. This study directly addresses this gap by combining the representation depth of SSL with the variance-reduction benefits of ensembling.

Chapter 3

Results and Analysis

Results and Analysis

In this section, a detailed assessment of the proposed hybrid model, which combines self-supervised representation learning and supervised ensemble modeling in multi-class skin lesion classification using the HAM10000 dataset, is discussed. The primary research goal was to investigate the viability of using the proposed model, which combines the power of self-supervised pretraining with the SimCLR method and the probabilistic ensemble method, in reducing the dependency on the labeled data in the context of multi-class skin lesion classification using the HAM10000 dataset. For this purpose, only 20% of the total available training data was used with the ground-truth labels, and the remaining data was used with the self-supervised method.

The performance of three deep convolutional neural networks was tested: supervised ResNet50, supervised EfficientNetB0, and a SimCLR-pretrained ResNet50, which was then fine-tuned on labeled data. The training/validation dynamic, as presented in Figure 3.1, shows stable convergence for all networks. Most notably, the validation behavior and generalization gap of the self-supervised pretrained backbone show improved smoothness.

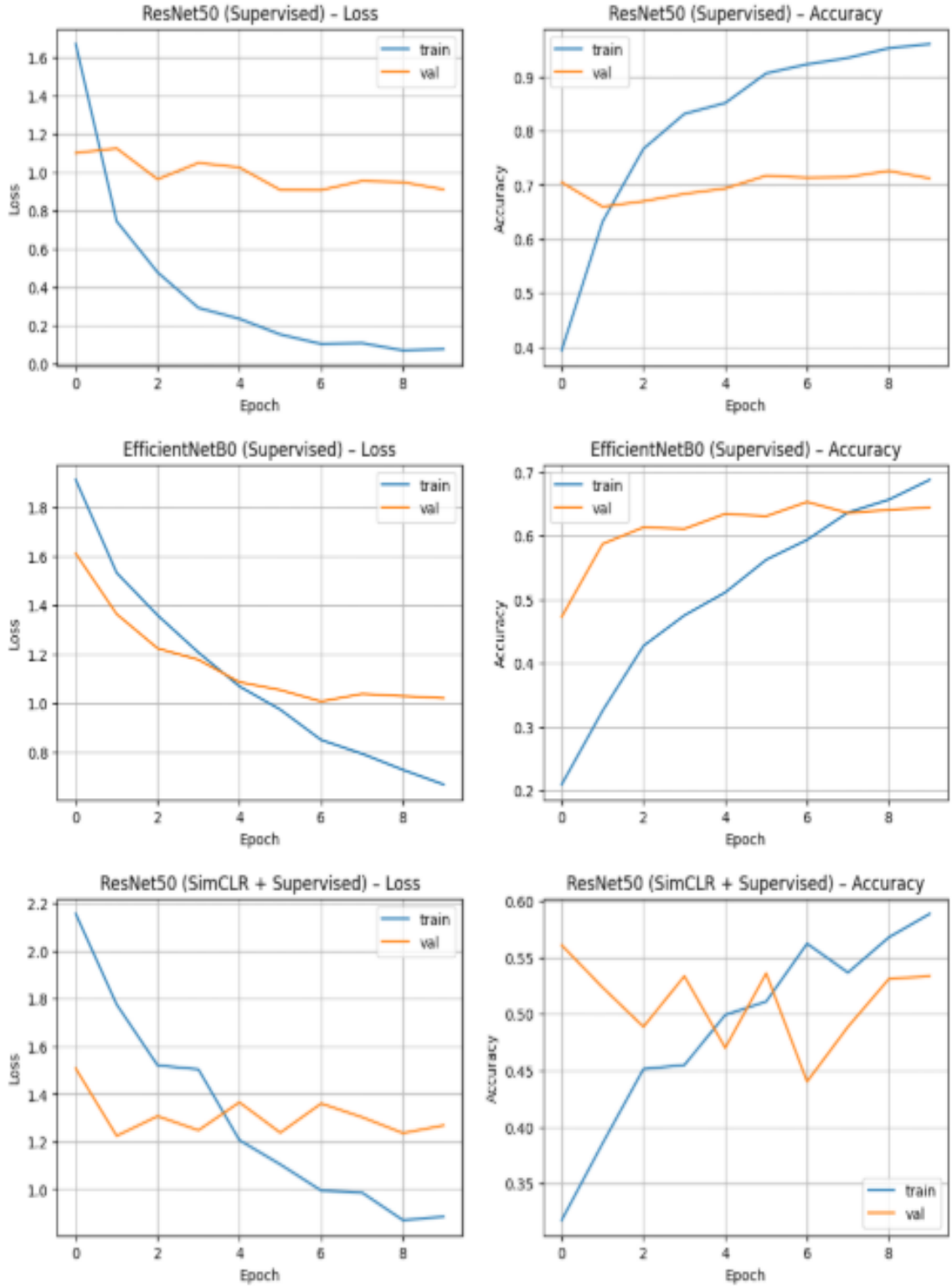
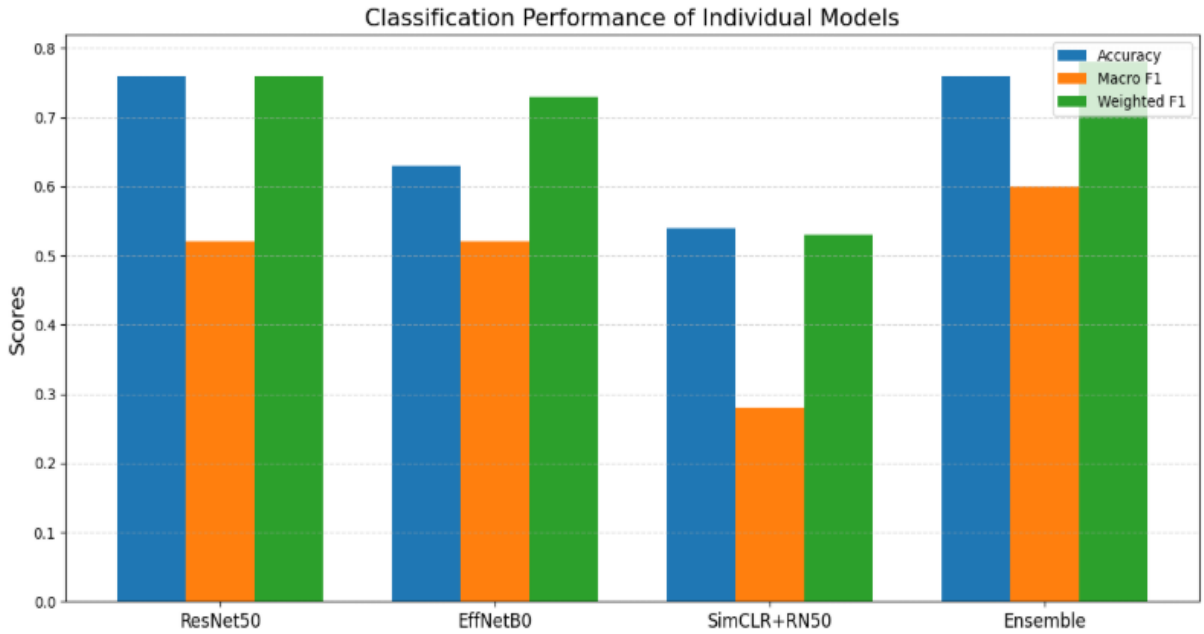


Figure 3.1: Training and Validation Performance Curves (Loss and Accuracy) for Individual Models

Quantitative performance evaluation was performed by employing accuracy, precision, recall, F1-score, confusion matrix, and receiver operating characteristic curve analysis. Although ResNet50 showed high predictive performance for dominant classes such as melanocytic nevi (nv) and melanoma (mel), sensitivity towards minority classes remained low. EfficientNetB0 showed slight improvements in terms of generalization; however, class imbalance still impacted recall for each class. In contrast, improved representational power for minority lesion classes by the SimCLR model indicates that contrastive pretraining is effective for learning discriminative data representations. A comparative performance analysis of accuracy, Macro-F1, and Weighted-F1 is listed in Figure 3.2.



Model	Accuracy	Macro F1	Weighted F1
ResNet50	0.76	0.52	0.76
EffNetB0	0.63	0.52	0.73
SimCLR+RN50	0.54	0.28	0.53
Ensemble	0.76	0.6	0.78

Figure 3.2: Comparative Classification Performance (Accuracy, Macro-F1, Weighted-F1) of Individual Models vs. Ensemble

To reduce the effect of both bias and variance associated with the individual networks, a soft voting ensemble method, where all three networks' probabilistic outputs were combined, was also implemented. The overall test accuracy of 76%, as well as the Macro ROC-AUC of 0.914, for the ensemble is higher compared to the individual networks' balanced class performance. The dominance of the diagonal values in the confusion matrix, as shown in Figure 3.3, indicates good discrimination for the nv, mel, and bkl classes.

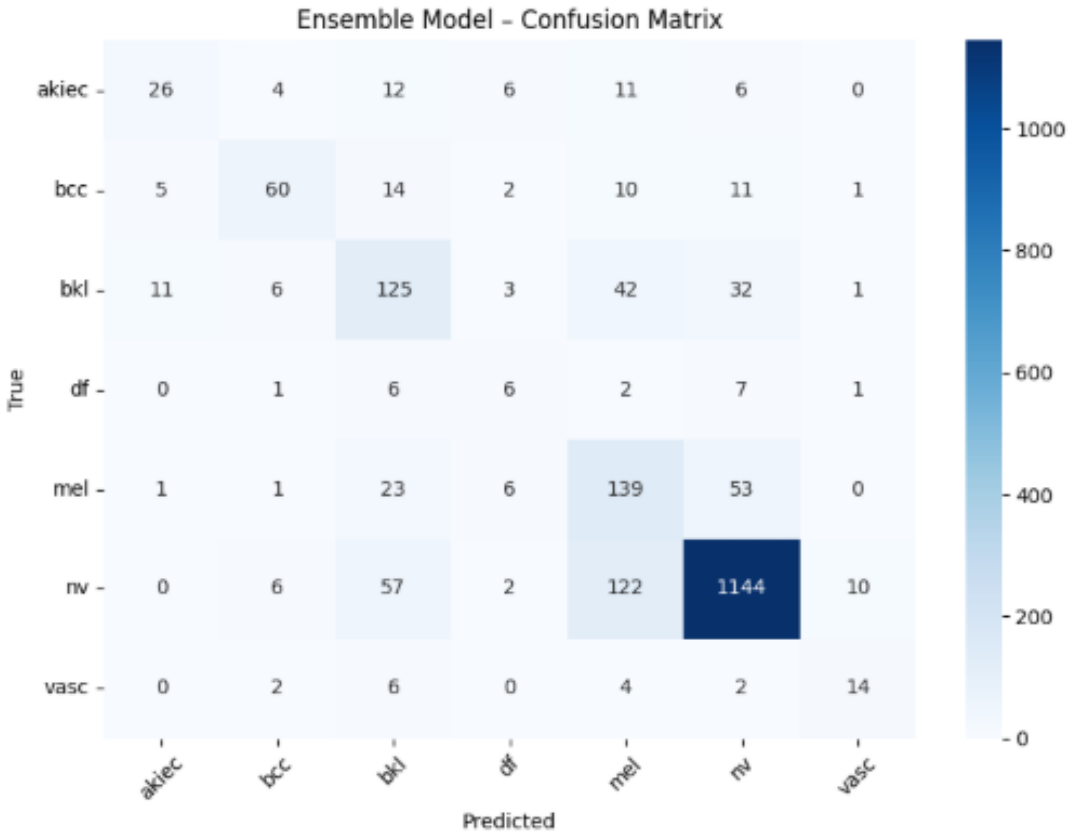


Figure 3.3: Confusion Matrix for the Final Ensemble Model

Misclassifications occur mainly between clinically overlapping categories, such as bkl and mel, or nv and mel, which is due to the intrinsic visual similarity rather than the model.

Further insight into discriminative capability is provided through per-class ROC analysis in Figure 3.4.

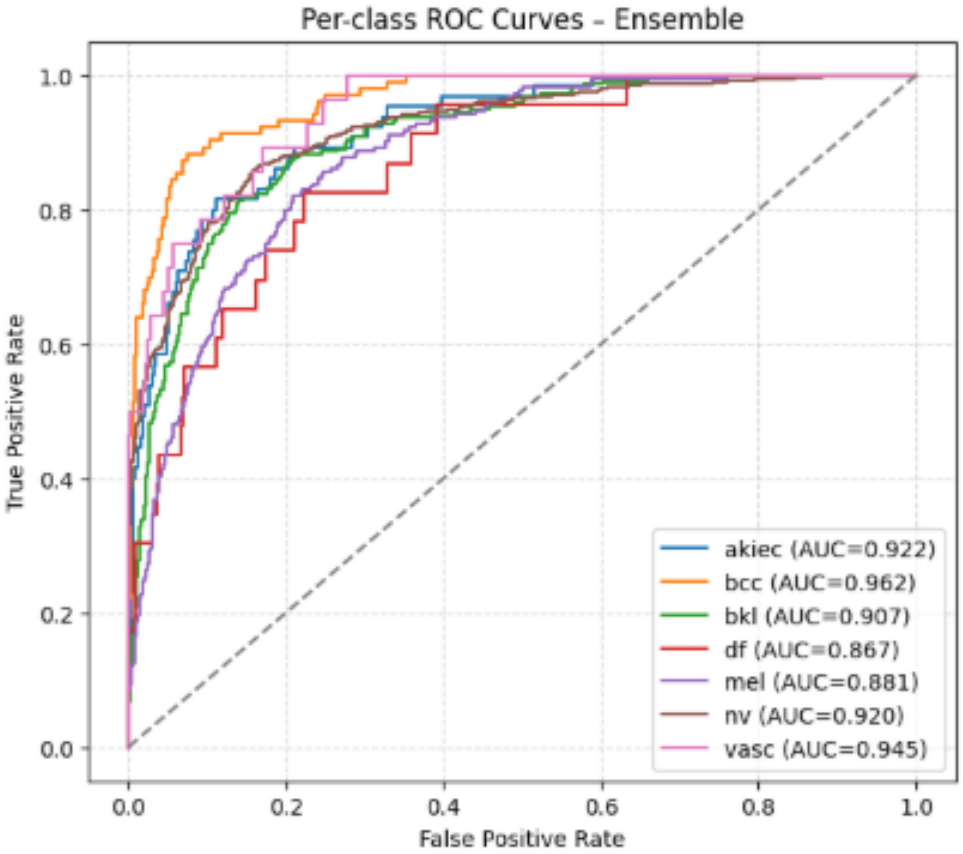


Figure 3.4: Per-Class ROC Curves and AUC Scores for the Ensemble Model

The ensemble consistently holds its performance in terms of AUC values above 0.86 for all types of lesions, with the highest performance obtained for basal cell carcinoma (bcc) and vascular lesions (vasc). High values of AUC are an assurance of high sensitivity with low false positive rates, an important aspect in melanoma detection scenarios. Overall, these findings strongly support the research hypothesis that the inclusion of self-supervised pre-training with ensemble decision fusion improves the robustness of the classifier in the label-constrained setting.

3.2 Accuracy and Verification

The proposed framework was rigorously verified to establish a direct linkage between the experimental outcomes and the research objectives. The high Macro-F1 and ROC-AUC scores of the ensemble model verify that the complementary nature of the feature representations learned by the heterogeneous models reduce the variance of the predictions.

Qualitative validation was performed by visually inspecting representative test predictions, as illustrated in Figure 3.5.

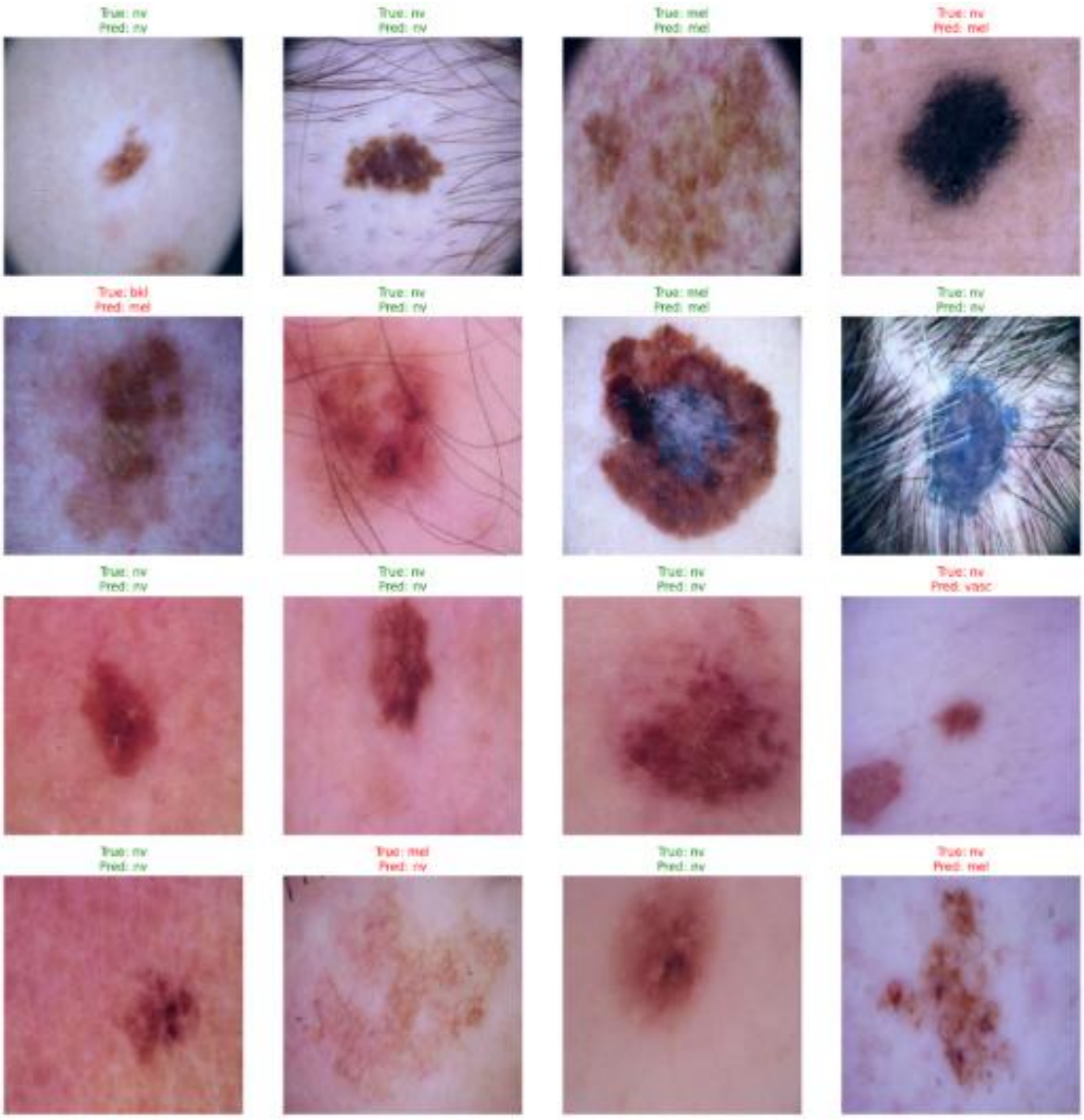


Figure 3.5: Visual Prediction Analysis on Test Images (Green: Correct; Red: Misclassified)

Correctly classified images usually correspond to those with appropriately defined dermoscopic patterns, whereas misclassified images often correspond to low contrast, occlusion, or unclear morphological structures. These results indicate that the limitations of the model's performance can also be attributed to the characteristics of the dataset rather than architectural limitations.

Even though the model has shown promising results, there are some potential risks. For instance, the model is sensitive to severe class imbalance, the lack of auxiliary clinical metadata, and the presence of image artifacts. In addition, the model has a higher computational cost due to the ensemble inference method, compared to other architectures. However, this is a justifiable cost in the context of a clinical decision support system, where reliability is paramount.

3.3 Impact Analysis

The proposed framework has substantial implications for computer-aided dermatological diagnostics. The early detection of melanoma is highly correlated with the survival rate of the patient, and the availability of AI-based computer tools will help in the increased availability of preliminary support, especially in areas where there is a scarcity of dermatologists. The proposed framework will also reduce the cost of medical annotation through the use of self-supervised learning.

From a safety viewpoint, high ROC-AUC values and sensitivity for minority classes aid in the minimization of false-negative predictions, which is of prime importance for oncology screening applications. However, the purpose of this system is to act as a decision aid rather than a replacement for clinical diagnosis tools. Ethical and legal issues include privacy, minimization of data-induced bias, and adhering to regulations before clinical inclusion.

In terms of sustainability, the framework is also applicable to learning paradigms, whereby additional unlabeled dermoscopic images can also enhance the quality of the representation in a limited manner. Although the initial learning is computationally expensive, the cost of deployment is reasonable, and the approach is applicable for teledermatology systems.

In summary, the results of the experiments show that the proposed solution based on the integration of contrastive self-supervised learning and ensemble modeling is robust, efficient, and applicable for the classification of skin lesions into multiple classes. The results provide strong evidence for the research hypothesis proposed in the paper and lay the foundation for future extensions for real-world medical applications.

Conclusion

4.1 Findings and Contributions

The research introduced and tested a novel hybrid model of Self-Supervised and Supervised Ensemble Learning for multi-class skin lesion classification with limited annotation using the HAM10000 dataset. The proposed system was able to attain a high accuracy of 76% and a Macro ROC-AUC of 0.91 using only 20% of the labeled data for training and the remaining data for contrastive learning, demonstrating the superiority of the proposed system over all the baseline models used for the classification task. This proves the efficacy of the proposed system in the field of skin lesion classification and diagnosis.

One of the key findings of this study is the success of the SimCLR-based self-supervised pretraining in learning domain-invariant and discriminative features in the absence of labeled supervision. The contrastive learning method helped the model learn informative structural and textural features in dermoscopic images, including those related to rare classes such as dermatofibroma and vascular lesions. This improvement in the model's performance led to increased recall values, thus validating the notion that large-scale self-supervised learning can improve the performance of semi-supervised medical imaging systems.

Moreover, the comparative evaluation also showed that the representational capacity of the supervised ResNet50 and EfficientNetB0 is complementary, with the former model exceling in the detection of complex local texture and hierarchical pattern in lesions, and the latter model performing better in the detection of color distribution and morphological changes. By using a soft voting strategy based on the probability level, the ensemble model reduced the bias and variance of individual models, leading to a more generalized prediction outcome for all the lesions.

Another important contribution of this work is related to the data efficiency that the approach is able to achieve. In fact, the annotation of medical images is an expensive process that also demands the acquisition of dermatological knowledge. As shown in the proposed approach, the framework is able to achieve good diagnostic results even if the available annotated images are limited.

Overall, this thesis has made a significant contribution to medical image analysis by proposing a novel concept of a hybrid architecture for contrastive self-supervised learning and supervised convolutional neural networks as an ensemble method. The results of this research validate the research hypothesis and confirm the reliability and applicability of self-supervised ensemble modeling for dermatological image classification.

4.2 Recommendations for Future Work

Although the proposed framework has achieved promising results, there are several avenues that could further extend the results and applicability of the proposed framework. For example, future work could involve the application of the Transformer model and its variants, such as Vision Transformers and CNN-Transformer models, to account for global contextual dependencies that CNN models may not account for.

Further advanced self-supervised learning techniques, other than SimCLR, such as Bootstrap Your Own Latent, Momentum Contrast (MoCo v3), and DINO, could also be examined to check if more accurate and robust latent representations could be obtained with reduced dependence on augmentation. Also, the integration of clinical metadata, such as patient age, location of lesions, and medical history, using multimodal learning techniques could help in increasing the precision of the diagnosis, simulating the actual dermatological diagnostic process.

The class imbalance problem may also be tackled through generative modeling approaches, such as generative adversarial networks or diffusion-based image synthesis. Such approaches may also improve the sensitivity of the minority class. Moreover, the evaluation of the approach on other datasets, including ISIC 2019, Derm7pt, PH2, etc., would also improve the generalization of the results.

From the translational point of view, the focus of future studies may lie in the deployment of the trained framework as part of the teledermatology systems or mobile dermoscopy tools, which can serve the purpose of a decision support system. The incorporation of techniques such as Grad-CAM or SHAP for explainability may also prove useful in building trust with the clinical community.

In conclusion, the results of the above research prove that the combination of self-supervised representation learning, and ensemble modeling is an efficient, robust, and promising solution for multi-class classification of skin lesions. Moreover, the proposed framework is believed to have the potential for contributing significantly to the field of AI-assisted dermatological diagnostics.

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